

Math 307 Project Overview and Instructions

The aim of the Math 307 project is to apply the topics studied in the Numerical Analysis course to analyze and solve various problems and/or case studies in any computational Sciences fields

Groups of **2-3 students** should propose and present a related idea as their final course project. Each group selects a topic to be addressed as a project. The topic idea should be communicated and approved by your instructor.

Project Milestones

Week	8th	10th	12th	14th
Task	• Form a team • Point/title • Short abstract	• Proposal • Analytical/ Numerical methodology	• Simulations • Codes • Draft report	• Presentation • Final report
% grade	10%	10%	10%	70%

Notes

- Form a team, select a topic and write an abstract **by week 8.** (10 points)
- Solve the proposed problem using numerical technique methods, compare with analytical method, if any, and write a proposal that contains abstract, introduction, methodology, and references about the employed technique **by week 10.** (10 points)
- Use ready-made SW packages such as Python, etc... for problem solving/simulating.
- Investigate the effect of the problem parameters on the behavior of the solution.
- Write a draft for the project report **by week 12.** (10 points)
- Submit the final report and give a presentation **by week 14** (70 points)